

SCHOLARPATH

- **Problem Statement Title-** ScholarPath—AI Agent for Citation Verification & Personalized Learning Roadmap
- **Theme-** EdTech / AI
- **PS Category-** Software
- **Team Name-** Error 404 Brain Not Found

IDEA TITLE

• Problem Statement

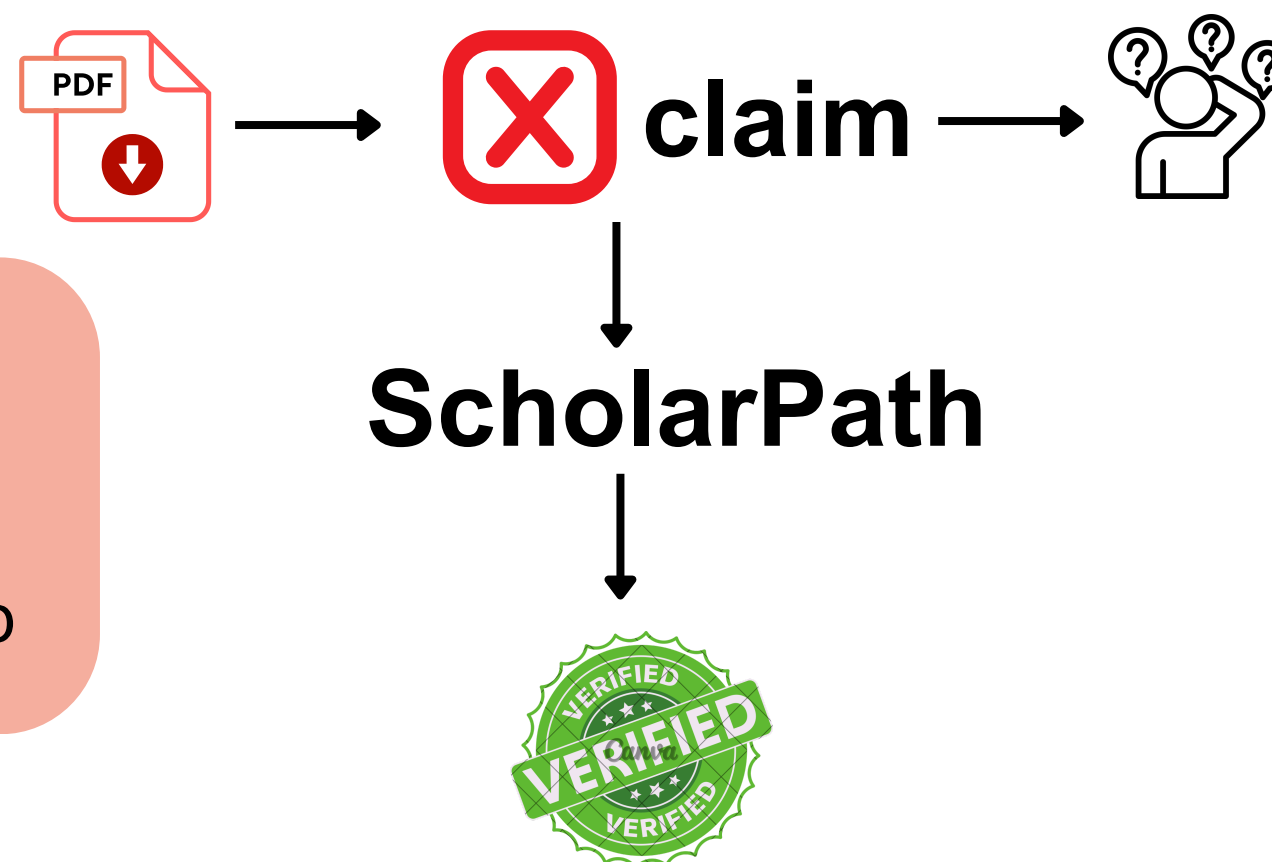
- ~25% citations are misrepresented
- No tool verifies academic claims before learning
- Students waste 10+ hrs/week on unreliable content

• Innovation

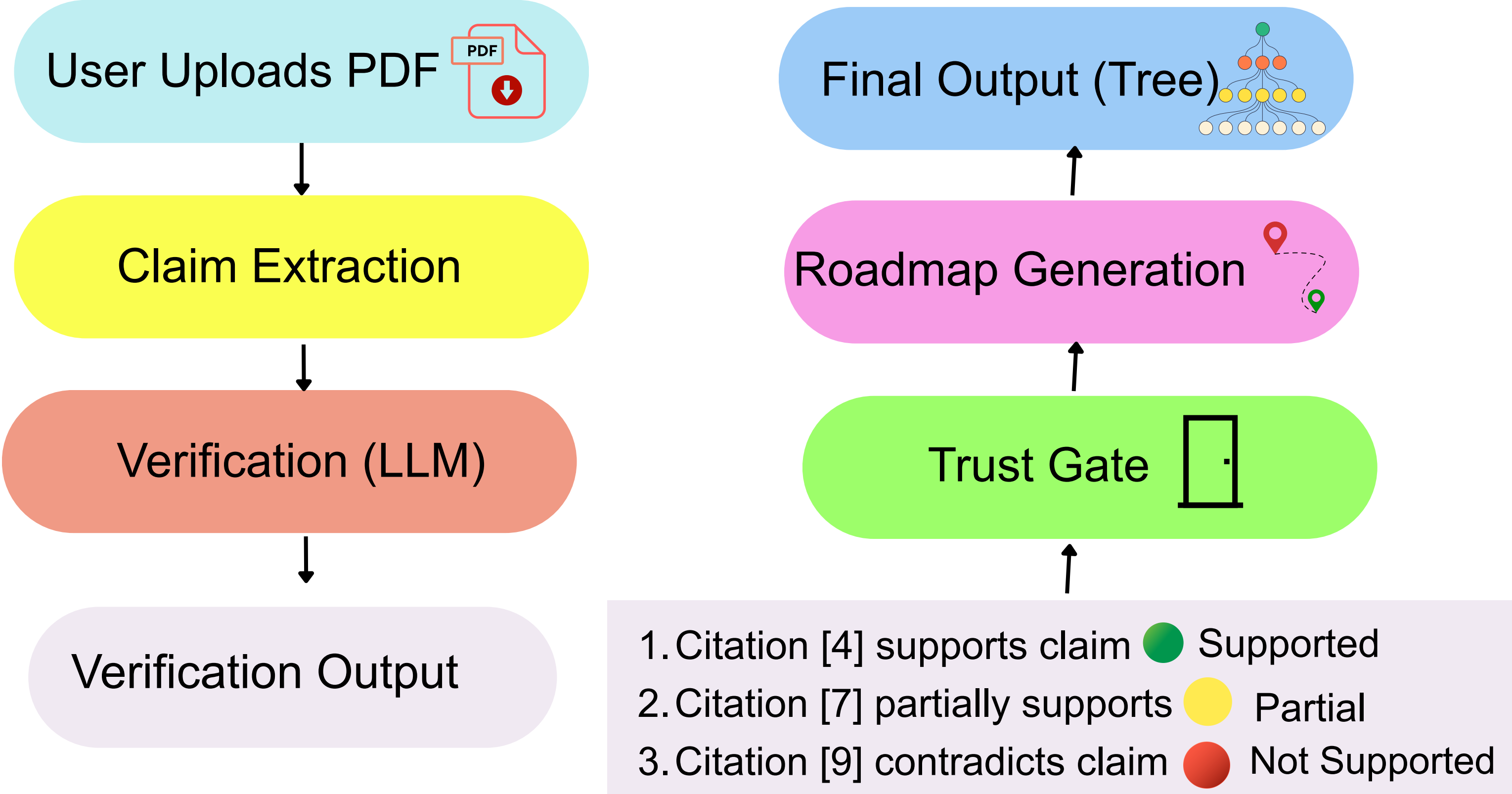
- Verify claim ↔ citation BEFORE learning
- Only trusted content generates roadmap
- Adapts to user's current knowledge

• Key Features

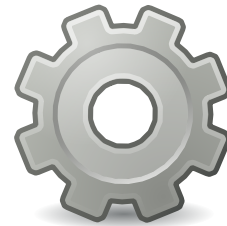
- Claim Extraction from PDF
- Citation Verification (AI)
- Trust Score Dashboard
- Personalized Learning Roadmap



ARCHITECTURE



• Tech Stack



- Backend: Python, FastAPI
- AI Pipeline: LangGraph (multi-agent)
- LLM: Ollama
- APIs: Semantic Scholar + arXiv
- Frontend: React.js + React Flow
- PDF Parsing: PyMuPDF

• Feasibility



- 200M+ papers accessible via APIs
- Works with open-access papers

• Challenges



- Paywalled papers → use abstracts
- LLM errors → confidence scoring

IMPACT AND REVENUE

• Impact

- Saves 10+ hrs per paper
- Prevents learning from misinformation
- Builds strong conceptual foundations

• Social Impact

- Helps students without mentors
- Especially impactful in India (low-quality PDFs issue)

• Revenue

- Freemium model (basic free, advanced insights paid)
- University integrations
- Research tools for scholars



Learn only from verified knowledge